# National Institute of Technology Sikkim

# Development of Online Students' Marksheet Generation and Distribution Management System

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Submitted to:
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# ⊡ Introduction

## What is the project about?

The project is a secure, web-based system to automate the entire marksheet lifecycle.

- **Purporse:**
  - To replace inefficient, manual process with a modern, digital solution
  - To provide a centralized platform for academic data management
- **Key Users:**
  - **Administrators:** For a system oversight and data management.
  - **Teachers:** For efficient marks submission.
  - **Students:** For secure, 24/7 access to their results
- **Core Benefits:**
  - Increases efficiency, ensures data accuraccy, and improves student satisfaction

# Problem Description

## Problems with existing manual systems:

**Inefficiency and Delays:**
Manual collection and calculation of marks is a slow and tedious process, costing both labour time and delays for overall operation.

**High Risk of Human Error:**
Human errors can creep up in marks and grade calculations if not done properly.

**Lack of Centralization:**
Data is scattered across physical files and spreadsheets, making it hard to manage and secure them.

**Limited Student Access & Privacy Concerns:**
- Students have to face time/location constraints to view results.
- Public noticeboards can compromise students' privacy which can lead to unwanted comparisons and eventual demotivation in students.

# System Architecture

- **Backend Framework: Django (Python)**
  Utilizes the Model-View-Template (MVT) architecture for a clean separation of concerns.
- **Database: MariaDB**
  A robust, open-source relational database for secure data storage.
- **Frontend:** HTML, CSS, and JavaScript
  Creates a responsive and user-friendly interface.
- **Key Libraries:**
  - **Pandas**: For processing spreadsheet uploads (.xlsx, .ods).
  - **ReportLab**: For generating dynamic, secure PDF documents.
  - **qrcode**: For creating data verification QR codes.

# Entities and Relations

**Problems with existing manual systems:**

**Core Entities:**

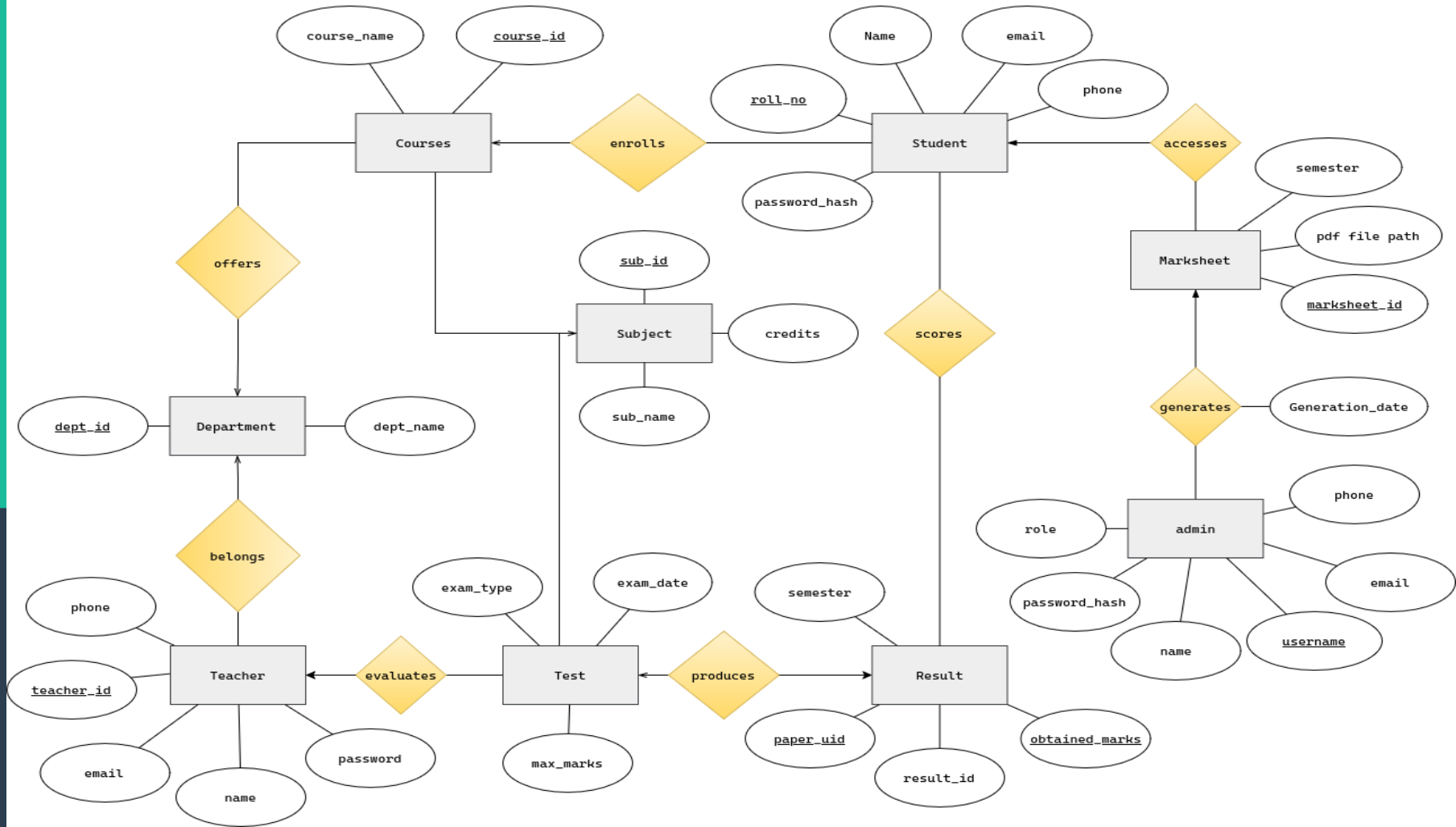
1) **Department**
2) **Course**
3) **Subject**
4) **Student**
5) **Teacher**
6) **Test**
7) **Result**
8) **Marksheet**

**Key Relationships:**

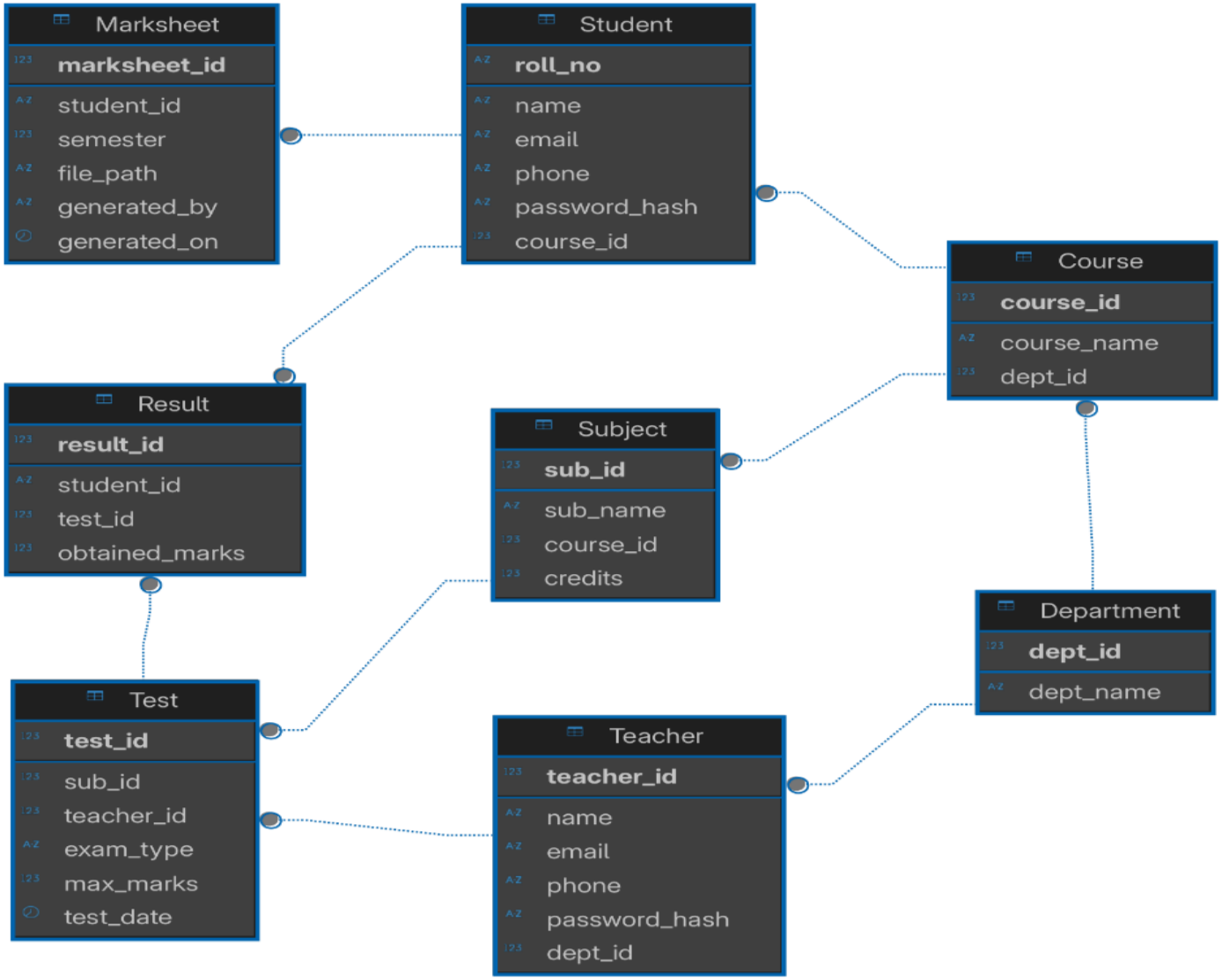
- A Department offers multiple Courses.
- A Course has many Subjects and enrolls many Students.
- A Teacher is assigned to conduct a Test for a Subject.
- A Student's performance in a Test is recorded as a Result.
- The system generates a Marksheet for a Student each semester, but only after an Admin publishes it.

# E-R Diagram

## ER Diagram



# Relational Schema



# ⚄ SQL Queries (via Django ORM)

**No Raw SQL**: We use Django's Object-Relational Mapper (ORM) for database operations.

Benefits of ORM:

- **Productivity**: Write database queries in Python, not SQL.
- **Security**: Automatically protects against SQL injection attacks.
- **Portability**: Easily switch databases (e.g., from MariaDB to PostgreSQL) with minimal code changes.

# ⚄ SQL Queries (via Django ORM)

**Example: Fetching a student's results for a subject**

**Python (ORM) Code:**

```
    Result.objects.filter(student__roll_no='B230038', test__subject__sub_code='CS15104')
```

**Equivalent SQL Generated by Django:**

```
SELECT * FROM academics_result
INNER JOIN academics_test ON academics_result.test_id = academics_test.id
WHERE academics_result.student_id = 'B230038'
AND academics_test.subject_id = (SELECT id FROM academics_subject WHERE
sub_code = 'CS15104');
```

# Relational Table

**Student Table:**

- roll_no (Primary Key)
- name, email, father_name, mother_name, reg_no
- course_id (Foreign Key to Course)
- user_id (Foreign Key to Django's User table)

**Subject Table:**

- id (Primary Key)
- sub_code (e.g., CS15104)
- sub_name, credits
- course_id (Foreign Key to Course)

**Result Table:**

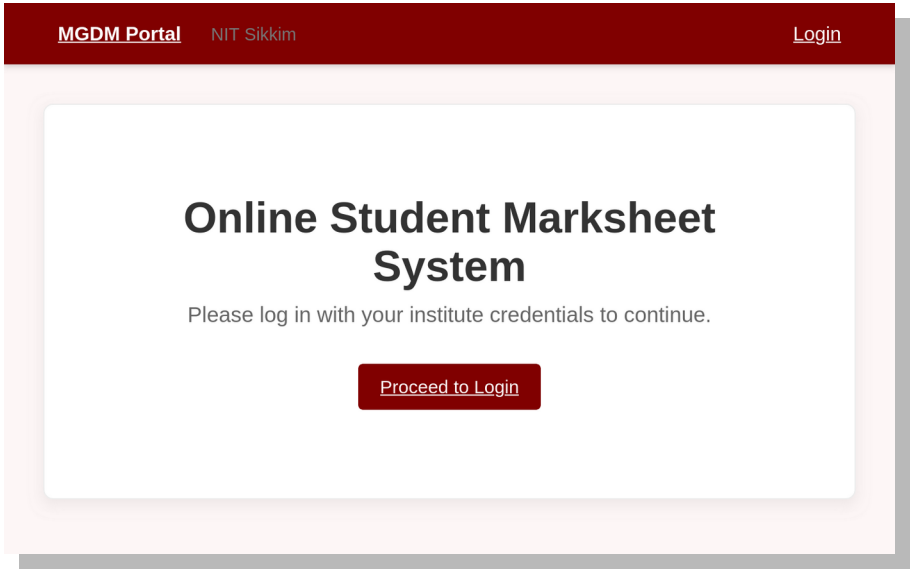
- id (Primary Key)
- obtained_marks
- student_id (Foreign Key to Student)
- test_id (Foreign Key to Test)

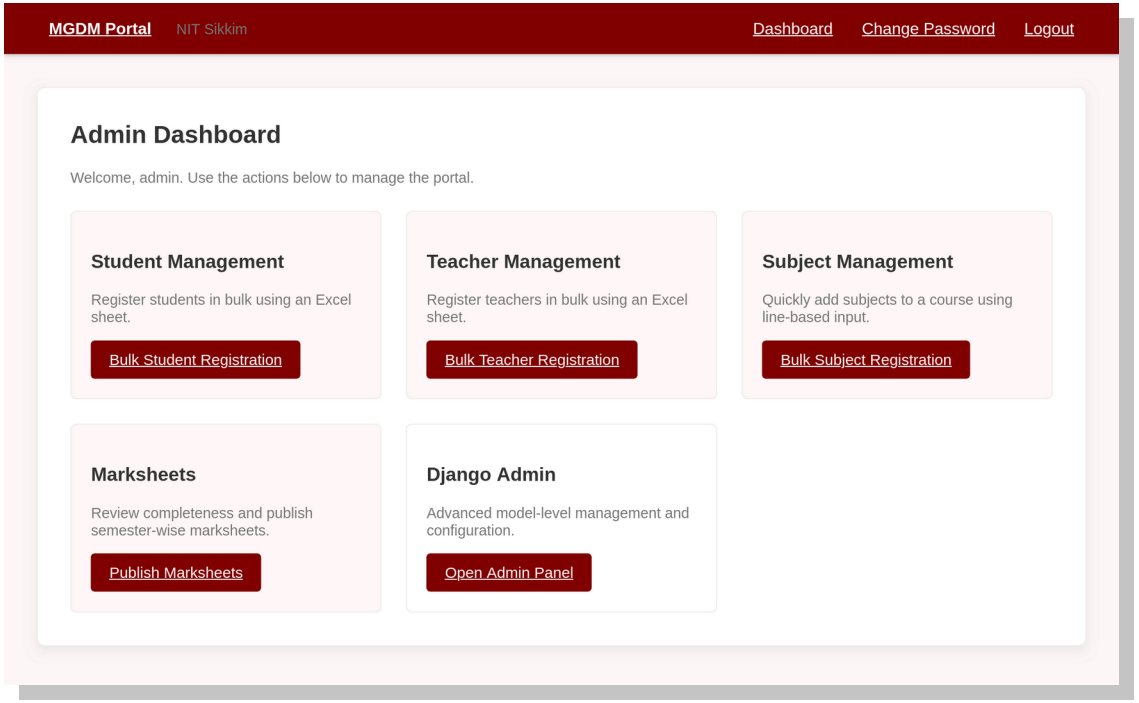
**Marksheet Table:**

- id (Primary Key)
- is_published (Boolean)
- student_id (Foreign Key to Student)
- semester

# ☰ Snapshots of the Project

## Home Page

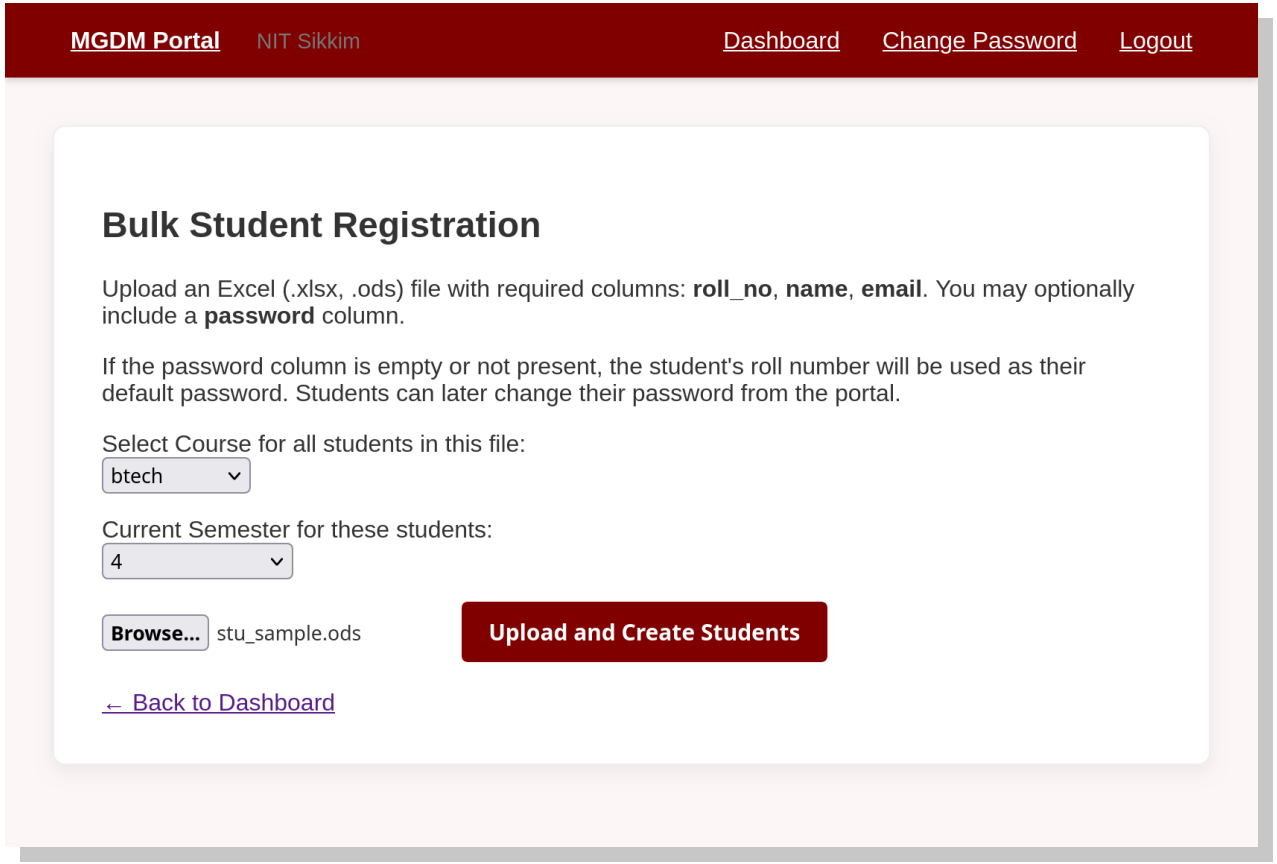



## Admin Dashboard

# Snapshots of the Project

**Bulk Student Registration Page**



# Django administration

WELCOME, **ADMIN**. VIEW SITE / CHANGE PASSWORD / LOG OUT

Home › Academics › Subjects › Add subject

Start typing to filter...

**ACADEMICS**

- Courses + Add
- Departments + Add
- Marksheets + Add
- Results + Add
- Students + Add
- Subjects + Add
- Teachers + Add
- Tests + Add

**AUTHENTICATION AND AUTHORIZATION**

- Groups + Add
- Users + Add

The subject "CS15101: Artificial Intelligence" was added successfully. You may add another subject below.

## Add subject

**Sub code:** CS15102

**Sub name:** Advanced Algorithms

**Course:** B.Tech CSE

**Credits:** 3

Teacher: Dr. Pratyay Kuila

SAVE | Save and add another | Save and continue editing

# ☰ Snapshots of the Project

## Teacher's View

MGDM Portal NIT Sikkim Dashboard Change Password Logout

### Publish Marksheets

Course: B.Tech CSE Semester: 5 Filter

#### Students in Selected Course & Semester

Tick students to publish/unpublish. Use the "View marks" link to verify detailed marks before publishing.

| Select | Student | Status | Marks completeness | Actions |
|---|---|---|---|---|
| ☐ | Sourav Mahato (B230032CS) | Not published | Missing marks in 7 subject(s) | View marks |
| ☐ | Sulav Singhal (B230033CS) | Not published | Missing marks in 7 subject(s) | View marks |
| ☐ | Sushil Gupta (B230034CS) | Not published | Missing marks in 7 subject(s) | View marks |
| ☐ | Tirthankar Ghosh (B230036CS) | Not published | Missing marks in 7 subject(s) | View marks |
| ☐ | Valle Durga Prasad (B230037CS) | Not published | Missing marks in 7 subject(s) | View marks |
| ☑ | Yash Gupta (B230038CS) | Not published | All marks present | View marks |

Publish Selected

MGDM Portal NIT Sikkim Dashboard Change Password Logout

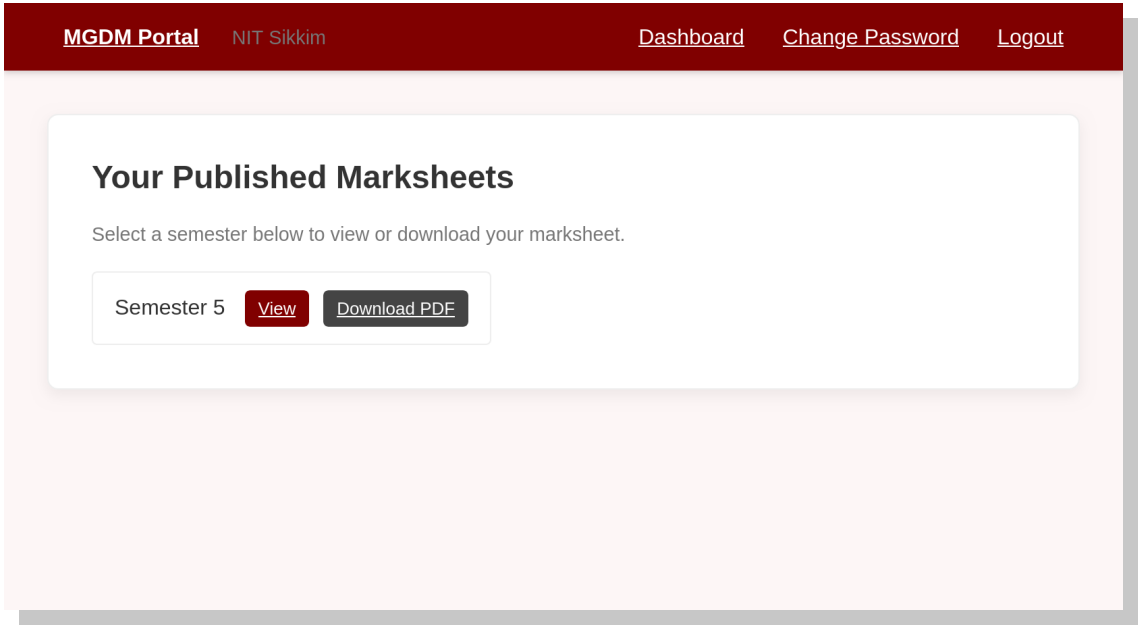
### Viewing Marks for: Artificial Intelligence (Midterm1)

| Roll No | Student Name | Marks Obtained | Max Marks | Edit |
|---|---|---|---|---|
| B230032CS | Sourav Mahato | 11 | 15 | Edit |
| B230033CS | Sulav Singhal | 11 | 15 | Edit |
| B230034CS | Sushil Gupta | 12 | 15 | Edit |
| B230036CS | Tirthankar Ghosh | 13 | 15 | Edit |
| B230037CS | Valle Durga Prasad | 13 | 15 | Edit |
| B230038CS | Yash Gupta | 11 | 15 | Edit |

← Back to Dashboard

# Snapshots of the Project

## Student's View



## PROVISIONAL GRADE CARD

Sign Out

### Student Information

| | | | |
|---|---|---|---|
| **Name:** | Yash Gupta | **Enroll No.:** | B230038CS |
| **Department:** | CSE | **Programme:** | B.Tech CSE |
| **Semester:** | 5 | | |

| Subject Name | Subject Code | Course Credit | Obtained Grade |
|---|---|---|---|
| Artificial Intelligence | CS15101 | 3 | AB |
| Advanced Algorithms | CS15102 | 3 | AB |
| Data Preprocessing and Mining | CS15103 | 3 | AB |
| Advanced Database Management Systems | CS15104 | 3 | AB |
| Artificial Intelligence Laboratory | CS15201 | 1 | AB |
| Advanced Algorithms Laboratory | CS15202 | 1 | SA |
| Data Preprocessing and Mining Laboratory | CS15203 | 1 | BB |
| Advanced Database Management Systems Laboratory | CS15204 | 1 | AA |

**SGPA: 8.12** **CGPA: 8.12** **Status: PASS**

← View Another Semester

Download PDF

*This result is intended solely for the examinee's immediate reference. For official use, please reach out to the Examination Cell, NIT Sikkim.*

Generated PDF Marksheet

# National Institute of Technology Sikkim

PROVISIONAL GRADE CARD

| | | | |
|---|---|---|---|
| **Name:** | Yash Gupta | **Enroll No.:** | B230038CS |
| **Department:** | CSE | **Programme:** | B.Tech CSE |
| **Semester:** | 5 | | |

| Subject Name | Subject Code | Course Credit | Obtained Grade |
|---|---|---|---|
| Artificial Intelligence | CS15101 | 3 | AB |
| Advanced Algorithms | CS15102 | 3 | AB |
| Data Preprocessing and Mining | CS15103 | 3 | AB |
| Advanced Database Management Systems | CS15104 | 3 | AB |
| Artificial Intelligence Laboratory | CS15201 | 1 | AB |
| Advanced Algorithms Laboratory | CS15202 | 1 | SA |
| Data Preprocessing and Mining Laboratory | CS15203 | 1 | BB |
| Advanced Database Management Systems Laboratory | CS15204 | 1 | AA |

**SGPA: 8.12 CGPA: 8.12 Status: PASS**

NIT Sikkim - B230038CS

# ☰ Conclusion

**Completion of Project:**

Successfully developed a fully functional, multi-user system for marksheet management.

Automated key process of marksheet generation and distribution, reducing manual effort and eliminating calculation errors.

Delivered a secure, efficient, and user-friendly platform for all stakeholders.

# ☷ Conclusion

**Future Scope:**

1. Transcript Generation: Add a feature to generate a consolidated transcript for all semesters.
2. Password Recovery: Implement a secure "Forgot Password" feature via email.
3. Analytics Dashboard: Provide administrators with visual analytics on student performance.

# Thankyou!!!